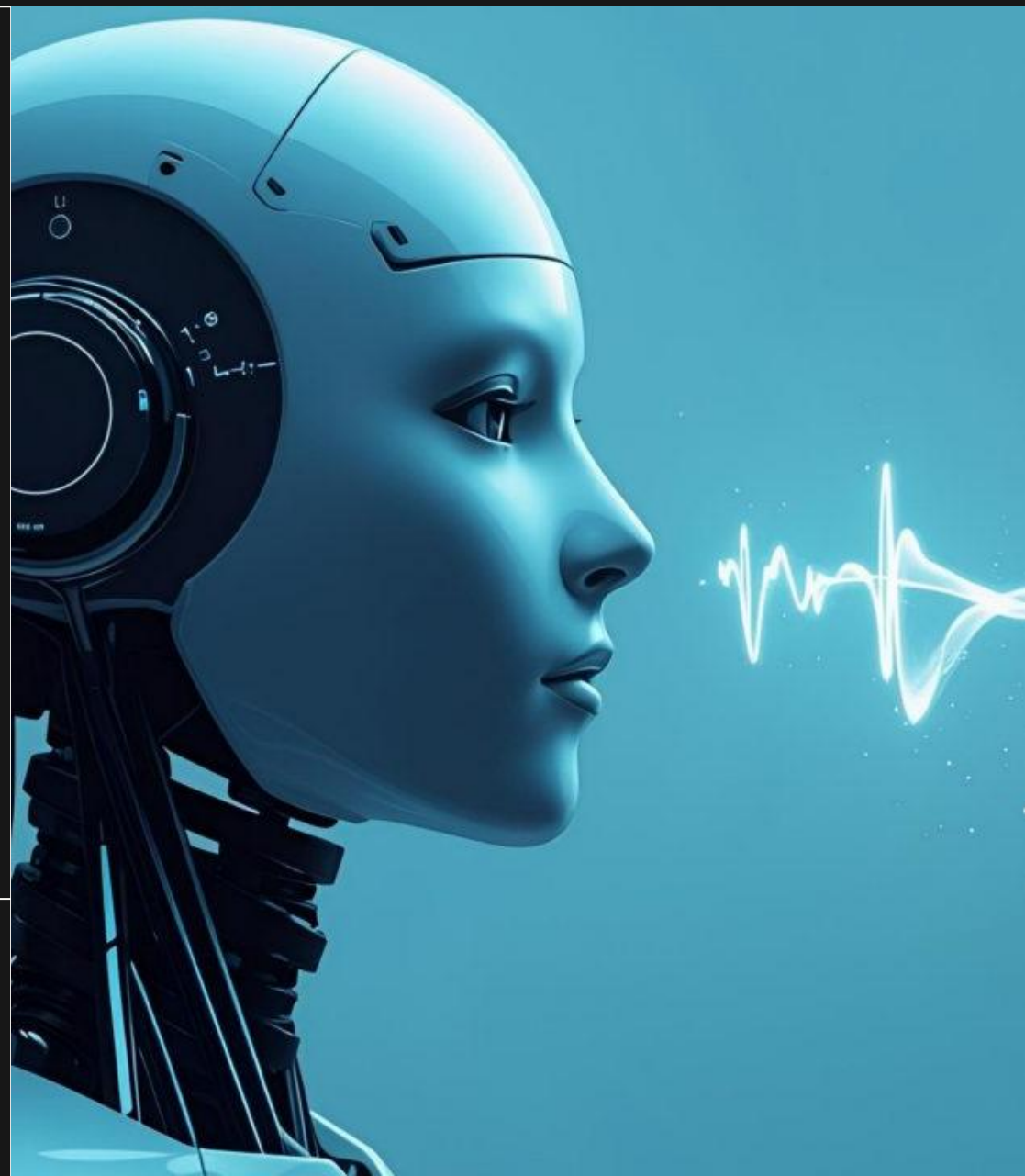
EVALUATION OF TWO AI APPLICATIONS: GOOGLE TRANSLATE & SIRI



MODULE: Found'ns of AI & Machine Learn

COURSE: COMPUTER SCIENCE



OVERVIEW OF AI IN EVERYDAY TOOLS

AI in Language Translation

AI tools like Google Translate use NLP and machine learning to provide accurate multilingual translation.

Voice-Activated AI Assistants

Voice assistants such as Siri leverage speech recognition and intelligent agent systems for user interaction.

AI Accessibility and Convenience

AI enhances accessibility and convenience for both specialist and non-specialist users in daily interactions.



WHAT IS GOOGLE TRANSLATE?

Multilingual Support

Google Translate is a multilingual translation service that converts text, speech, images, and websites to over 130 languages, enabling communication across diverse linguistic groups.

Who developed the application?

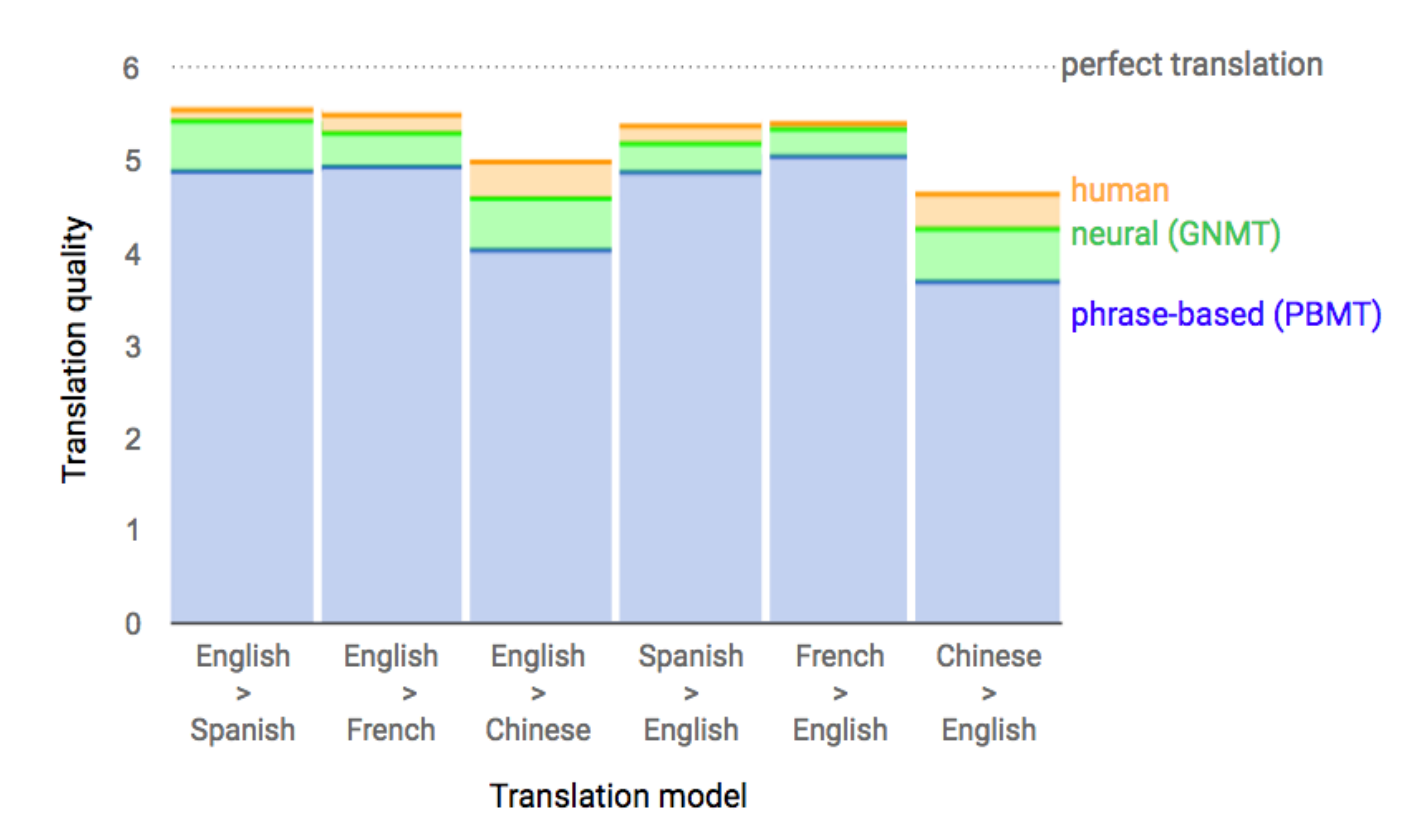
Google (a subsidiary OF Alphabet Inc.), was originally launched in April 2006, as a research project at Google, led by Franz Josef Och, using statistical machine translation to generate translations from large bilingual datasets.

Accessibility and Usage

Google Translate is free and widely accessible, popular among individuals and businesses worldwide.

AI TECHNOLOGIES USED IN GOOGLE TRANSLATE

- Neural Machine Translation
  - NMT enables accurate, context-aware translations by using deep neural networks to understand language nuances.
- Natural Language Processing
  - NLP helps in analyzing and understanding linguistic structures for better translation quality.
- Deep Learning Enhancement
  - Deep learning techniques continually improve translation accuracy by learning from vast data.
- Computer Vision for Text
  - Computer vision allows text recognition and translation within images for versatile language support.



Data from side-by-side evaluations, where human raters compare the quality of translations for a given source sentence. Scores range from 0 to 6, with 0 meaning "completely nonsense translation", and 6 meaning "perfect translation."

BENEFITS, ETHICAL ISSUES & IMPROVEMENTS

Benefits of Translation Technology

- Breaks language barriers globally
- Free and accessible on web and mobile
- Real-time translation for text, speech, and images
- Helps travellers, students, and businesses communicate
- Supports low-literacy users through voice and camera translation

Ethical Issues

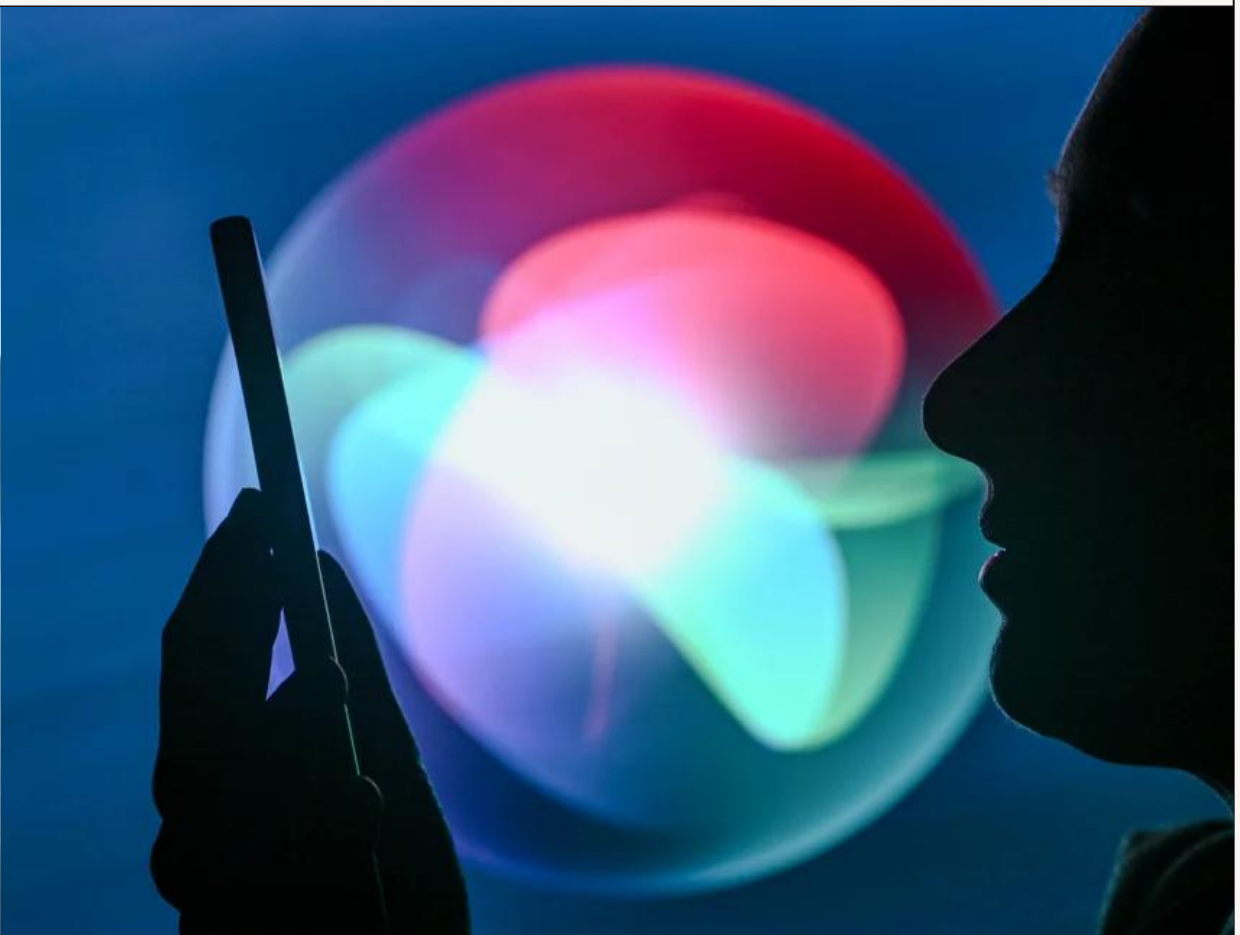
- Inaccurate translations may cause misunderstandings
- Bias in training data can lead to culturally insensitive outputs
- Privacy concerns when users translate sensitive text
- Over-reliance may reduce language-learning motivation

Suggested Improvements

Enhance contextual understanding, support low-resource languages, and increase data handling transparency.



WHAT IS SIRI?



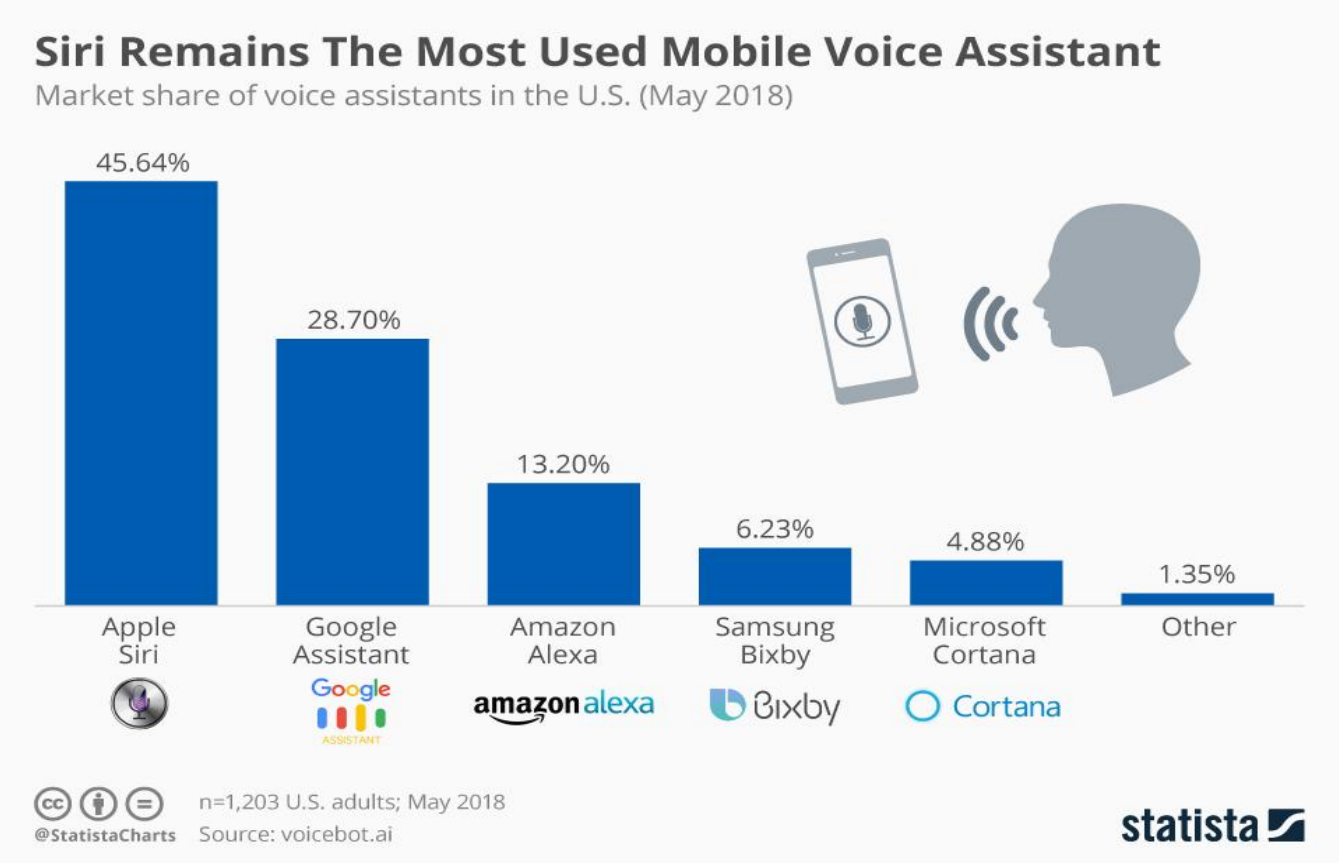
Voice-Activated Assistant

Siri is a voice-activated assistant that allows hands-free interaction with devices using natural language. Siri performs tasks like answering questions, controlling apps, and managing device functions efficiently.

Who developed the application?

Siri originated as a DARPA-funded AI project at SRI International, spun off into a startup in 2007 by Dag Kittlaus, Tom Gruber, and Adam Cheyer, then acquired by Apple in 2010, becoming a built-in iPhone feature in 2011 released with the iPhone 4S. The name, chosen by Kittlaus from his Norwegian heritage, is short for Sigrid, meaning "beautiful woman who leads you to victory"

AI TECHNOLOGIES USED IN SIRI



Apple's Siri leads the U.S. market for voice assistants on mobile phones with 45.65 percent. Even though available on most Android devices, the Google Assistant only has a market share of 28.9 percent while Samsung's Bixby even ranks below Amazon's Alexa, a voice assistant which is not native to any Smartphone.

Speech Recognition

Converts spoken language into text to understand user commands accurately and efficiently.

Natural Language Processing

Interprets user intent and context to enable meaningful and relevant responses.

Machine Learning

Learns from user behavior to improve response accuracy and personalization over time.

Intelligent Agent Systems

Automates task execution like reminders, messaging, and web searches for user convenience.

Deep Learning (Neural Networks)

Used in the speech recognition engine for sophisticated voice understanding.

BENEFITS, ETHICAL ISSUES & IMPROVEMENTS



Key Benefits

- Hands-free convenience for everyday tasks
- Accessibility support for users with disabilities
- Integrates with smart home devices
- Learns user preferences over time
- Helps with reminders, navigation, messaging, and more

Ethical Concerns

- Privacy concerns due to voice data collection
- Risk of accidental activation (“Hey Siri” triggers)
- Bias in speech recognition (accents, dialects)
- Cloud-based processing raises security concerns
- Potential over-reliance on automated decision-making

Suggested Improvements

Enhancing accent recognition, increasing on-device processing to protect privacy, and improving contextual awareness for complex tasks are key focus areas.



SUMMARY AND KEY TAKEAWAYS

AI Enhances Communication

AI tools like translation services enable global communication but need better contextual accuracy.

Virtual Assistant Benefits

Virtual assistants provide hands-free help and personalized user experiences.

Privacy and Ethical Challenges

AI development must address privacy, recognition errors, and uphold ethical standards.

Continuous Improvement

Ongoing refinement ensures AI technologies remain trustworthy and beneficial.